CKD — Definition & KDIGO Staging

1. Over 3 months banner

WHAT YOU SEE

Castle banner reading 'CKD over 3 months'.

WHAT IT ENCODES

CKD = abnormalities of kidney STRUCTURE or FUNCTION present for >3 MONTHS with implications for health. Operationally: eGFR <60 mL/min/1.73m2, OR markers of kidney damage (albuminuria ≥30 mg/g, abnormal urine sediment/hematuria, electrolyte/tubular disorders, structural abnormalities on imaging, biopsy abnormalities, or transplant history) — EITHER persisting >3 months. Chronicity (>3 months) is exactly what separates CKD from AKI. Mnemonic: '<60 OR damage marker, for >3 months.'

BOARD TRAP

Stem: a single lab shows eGFR 55 on one visit — wrong answer is to immediately diagnose CKD (or to label a 3-day creatinine rise as CKD). Discriminator: CKD requires PERSISTENCE >3 months; a one-time low eGFR or a single abnormal value may be AKI or transient — you must confirm chronicity (repeat testing) before diagnosing CKD.

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2. Detective / occult FSGS

WHAT YOU SEE

Detective with magnifying glass and a 'WANTED' poster naming HTN-CKD and occult FSGS.

WHAT IT ENCODES

Always investigate the CAUSE of CKD rather than reflexively labeling it 'hypertensive' or 'diabetic.' Occult primary glomerular disease — especially FSGS (focal segmental glomerulosclerosis, often with heavy proteinuria) and other glomerulonephritides — is commonly missed. Clues to look for an alternate cause: significant proteinuria, hematuria/dysmorphic RBCs or casts, rapid decline, or CKD without diabetic retinopathy in a 'presumed diabetic.' Mnemonic: 'don't blindly accept the label — hunt the real culprit.'

BOARD TRAP

Stem: diabetic with CKD, heavy proteinuria, hematuria, but NO retinopathy — wrong answer is to assume diabetic nephropathy and stop. Discriminator: in diabetes, nephropathy almost always coexists with RETINOPATHY; its ABSENCE (or active sediment/hematuria, very rapid decline, or proteinuria out of proportion) should prompt biopsy to find FSGS or another glomerulonephritis rather than attributing all CKD to diabetes/HTN.

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3. KDIGO eGFR staircase (G1–G5)

WHAT YOU SEE

Green-to-red staircase listing G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15.

WHAT IT ENCODES

KDIGO GFR categories (mL/min/1.73m2): G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 (or dialysis = G5D). Splitting G3 into 3a/3b matters because risk and management differ. Lower number = lower GFR = 'redder' = worse. Mnemonic: count down by halving-ish — 90, 60, 45, 30, 15.

BOARD TRAP

Stem: healthy young person, eGFR 85 (G2), no proteinuria, no hematuria, normal imaging — wrong answer is to diagnose CKD because eGFR is 'below 90.' Discriminator: G1 (≥90) and G2 (60–89) are NOT CKD by themselves — they require a co-existing DAMAGE MARKER (albuminuria, hematuria, structural/biopsy abnormality). Only eGFR <60 (G3a+) qualifies as CKD on GFR alone.

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4. Albuminuria A1/A2/A3

WHAT YOU SEE

Color tabs A1, A2, A3 next to the staging castle.

WHAT IT ENCODES

Albuminuria categories by urine albumin-to-creatinine ratio (ACR, mg/g): A1 <30 (normal/mildly increased), A2 30–300 (moderately increased, the old 'microalbuminuria'), A3 >300 (severely increased, old 'macroalbuminuria'). KDIGO stages risk by BOTH eGFR (G) AND albuminuria (A) — the green/yellow/orange/red 'heat map.' Mnemonic: 'CGA' = Cause + GFR category + Albuminuria category.

BOARD TRAP

Stem: two patients both at G3a — one A1, one A3 — treated as identical risk — wrong answer. Discriminator: staging on eGFR ALONE misses risk; albuminuria is an INDEPENDENT predictor of CKD progression AND cardiovascular death, so the A3 patient (heat-map red) is far higher risk and warrants more aggressive RAAS blockade/SGLT2i and monitoring. Always report G AND A together.

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5. Global burden rising / world map

WHAT YOU SEE

World map labeled 'global burden rising'.

WHAT IT ENCODES

CKD prevalence (~10% of adults globally) is rising worldwide, driven mainly by the diabetes and hypertension epidemics plus aging. CKD is a leading and growing cause of death, and a major cost/dialysis burden. Mnemonic: 'diabetes + hypertension + aging = a global CKD wave.'

BOARD TRAP

Stem framing CKD as a rare/declining problem or attributing the global rise primarily to glomerulonephritis/infections — wrong answer. Discriminator: the global rise is driven predominantly by DIABETES and HYPERTENSION (lifestyle/metabolic disease) and aging, not by primary glomerular or infectious causes.

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6. King — etiology directed

WHAT YOU SEE

Crowned king at castle gate labeled 'etiology-directed'.

WHAT IT ENCODES

Management has TWO arms: (1) treat the underlying CAUSE (glycemic control in diabetes, BP control, immunosuppression for glomerulonephritis, relieve obstruction, etc.), and (2) generic kidney-protective measures (RAAS blockade, SGLT2i, BP/proteinuria targets, avoid nephrotoxins, treat CKD complications). The KDIGO 'C' (Cause) is part of formal staging. Mnemonic: 'name the cause, then protect the nephron.'

BOARD TRAP

Stem: CKD patient given only generic renoprotection while an obstructing stone / treatable glomerulonephritis / RAS goes unaddressed — wrong answer. Discriminator: management must be ETIOLOGY-DIRECTED — generic measures don't substitute for treating a reversible/specific cause (relieve obstruction, immunosuppress active GN, control diabetes); missing the cause is the error.

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7. RAAS vicious cycle wheel

WHAT YOU SEE

Mill water-wheel labeled 'RAAS vicious cycle'.

WHAT IT ENCODES

Nephron loss activates the renin-angiotensin-aldosterone system; angiotensin II constricts the EFFERENT arteriole, raising intraglomerular pressure to maintain GFR — but this glomerular hypertension and aldosterone-driven fibrosis progressively destroy remaining nephrons (a self-perpetuating cycle). This is the rationale for ACEi/ARB (and SGLT2i, which restore tubuloglomerular feedback). Mnemonic: the wheel that 'turns nephron loss into more nephron loss.'

BOARD TRAP

Stem: patient with normal systemic BP but progressive proteinuric CKD — wrong answer is to assume the glomerulus is fine because the cuff pressure is normal. Discriminator: INTRAGLOMERULAR pressure (efferent constriction from angiotensin II) can stay high and damage the glomerulus even when SYSTEMIC BP is normal — which is why RAAS blockade slows progression independent of BP, and why proteinuria persists despite normal cuff readings.

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8. Hyperfiltration tower

WHAT YOU SEE

Tower labeled 'hyperfiltration tower' with workers.

WHAT IT ENCODES

After nephron loss, surviving nephrons COMPENSATE by hyperfiltering (afferent dilation + efferent constriction → single-nephron hyperfiltration). This maintains total GFR short-term but causes shear stress, glomerular hypertrophy/sclerosis, and proteinuria — accelerating destruction of the remaining nephrons (the 'final common pathway' of progression). Same mechanism underlies early diabetic hyperfiltration. Mnemonic: 'the survivors overwork and burn out.'

BOARD TRAP

Stem: early diabetic with an ELEVATED eGFR (hyperfiltration) — wrong answer is to call the kidneys 'extra healthy' or low-risk. Discriminator: hyperfiltration is a MALADAPTIVE early warning of nephron stress/progression, not robust health; it precedes albuminuria and decline. SGLT2i/RAAS blockade reduce it — high eGFR here is a red flag, not reassurance.

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9. ACEi/ARB knight

WHAT YOU SEE

Knight with shield labeled 'ACE/ARB'.

WHAT IT ENCODES

ACEi/ARB are first-line to SLOW CKD progression, with the greatest benefit in ALBUMINURIC CKD (diabetic and non-diabetic). They lower intraglomerular pressure (efferent dilation) → reduce proteinuria and nephron loss. Titrate to the maximum tolerated dose; monitor creatinine and potassium 1–2 weeks after starting/up-titration. Mnemonic: the 'shield' that drops glomerular pressure and proteinuria.

BOARD TRAP

Stem: ACEi started for albuminuric CKD, creatinine rises ~25% — wrong answer is to discontinue it. Discriminator: an early creatinine rise UP TO ~30% is EXPECTED and acceptable (reflects intended lowering of glomerular pressure) — continue and monitor. Stop and investigate (e.g., bilateral renal artery stenosis, volume depletion) only if the rise exceeds ~30% or potassium becomes dangerous; and never combine ACEi + ARB.

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10. SGLT2 shield

WHAT YOU SEE

Small shield labeled 'SGLT2'.

WHAT IT ENCODES

SGLT2 inhibitors (dapagliflozin, empagliflozin) slow CKD progression and reduce CV/renal events by restoring tubuloglomerular feedback (afferent constriction → lower intraglomerular pressure) plus natriuresis and metabolic effects. Proven in DIABETIC AND NON-DIABETIC proteinuric CKD (DAPA-CKD, EMPA-KIDNEY); cause a small, expected, reversible early eGFR dip. Mnemonic: the 'second shield' added on top of ACEi/ARB for cardio-renal protection.

BOARD TRAP

Stem: non-diabetic CKD patient with proteinuria already on a maxed ACEi — wrong answer is to withhold an SGLT2 inhibitor because the patient 'isn't diabetic,' or to stop it over the early eGFR dip. Discriminator: SGLT2i benefit is NOT limited to diabetics (proven in non-diabetic CKD) and is ADDITIVE to RAAS blockade; the small initial eGFR dip is hemodynamic and expected, not a reason to stop.

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11. Etiology gallery (DN, GN, HTN, ADPKD)

WHAT YOU SEE

Row of monster bottles: diabetic nephropathy, glomerulonephritis, hypertension-CKD, ADPKD/hereditary.

WHAT IT ENCODES

Leading causes of CKD/ESRD: diabetic nephropathy is #1 (most common cause of ESRD overall), then hypertensive nephrosclerosis, glomerulonephritis, and ADPKD/hereditary disease. Diabetic nephropathy classically progresses hyperfiltration → microalbuminuria → macroalbuminuria → declining GFR, and usually co-occurs with retinopathy. Mnemonic: 'Diabetes Dominates' the ESRD list.

BOARD TRAP

Stem asks the single most common cause of ESRD — wrong answer is glomerulonephritis or hypertension. Discriminator: DIABETIC NEPHROPATHY is the most common cause of ESRD overall (hypertension is second). Don't pick GN or HTN as #1; and remember ADPKD is the most common HEREDITARY cause.

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12. Risk factor icons

WHAT YOU SEE

Icons: smoking, obesity, CV disease, proteinuria, prematurity/low birth weight.

WHAT IT ENCODES

CKD risk factors include smoking, obesity, cardiovascular disease, proteinuria/albuminuria, diabetes, hypertension, AKI episodes, nephrotoxin exposure, family history, and reduced nephron endowment from PREMATURITY/LOW BIRTH WEIGHT (fewer nephrons from birth → lifelong vulnerability). Many are modifiable. Mnemonic: 'fewer nephrons at birth = more CKD later.'

BOARD TRAP

Stem: young adult born premature/low birth weight now with early proteinuria/HTN — wrong answer is to dismiss CKD risk as unrelated to birth history. Discriminator: low nephron endowment (prematurity/LBW) is an established, often-overlooked CKD risk factor (low congenital nephron number → hyperfiltration and earlier CKD); combine with modifiable risks (smoking, obesity, proteinuria) for risk assessment.

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13. Race-free eGFR equation

WHAT YOU SEE

Scale/equation icon labeled 'race-free equation'.

WHAT IT ENCODES

Current practice uses the 2021 CKD-EPI creatinine equation WITHOUT the race coefficient (race is a social, not biological, variable). When creatinine-based eGFR is unreliable (extremes of muscle mass — amputees, bodybuilders, cachexia, cirrhosis, paraplegia), CYSTATIN C (or combined creatinine–cystatin C) refines the estimate. Mnemonic: 'no race coefficient; use cystatin C when muscle mass is weird.'

BOARD TRAP

Stem: low-muscle-mass cachectic patient with a 'normal' creatinine — wrong answer is to trust the creatinine-based eGFR as accurate (or to apply the old race adjustment). Discriminator: low muscle mass produces a falsely LOW creatinine → falsely HIGH/reassuring eGFR; use CYSTATIN C-based eGFR for a truer estimate. And never apply the obsolete race coefficient.

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14. Uremia brain (sequela)

WHAT YOU SEE

Brain labeled 'uremia' top right.

WHAT IT ENCODES

Advanced CKD (G5, eGFR <15) produces the UREMIC SYNDROME — encephalopathy/asterixis, pericarditis, nausea/anorexia, platelet dysfunction/bleeding, neuropathy, pruritus — from retained toxins. Symptomatic uremia is a key indication to start dialysis (or transplant). Mnemonic: dialysis indications = AEIOU — Acidosis, Electrolytes (refractory hyperkalemia), Intoxications, Overload (refractory), Uremic symptoms (pericarditis/encephalopathy/bleeding).

BOARD TRAP

Stem: asymptomatic patient at eGFR 13 with normal electrolytes — wrong answer is to start dialysis purely because the eGFR number crossed a threshold. Discriminator: dialysis initiation is driven by SYMPTOMS/complications (AEIOU — uremic pericarditis, encephalopathy, refractory hyperkalemia/acidosis/overload, intoxication), not by an eGFR number alone; trials (IDEAL) show no benefit to early, asymptomatic 'eGFR-triggered' starts.
